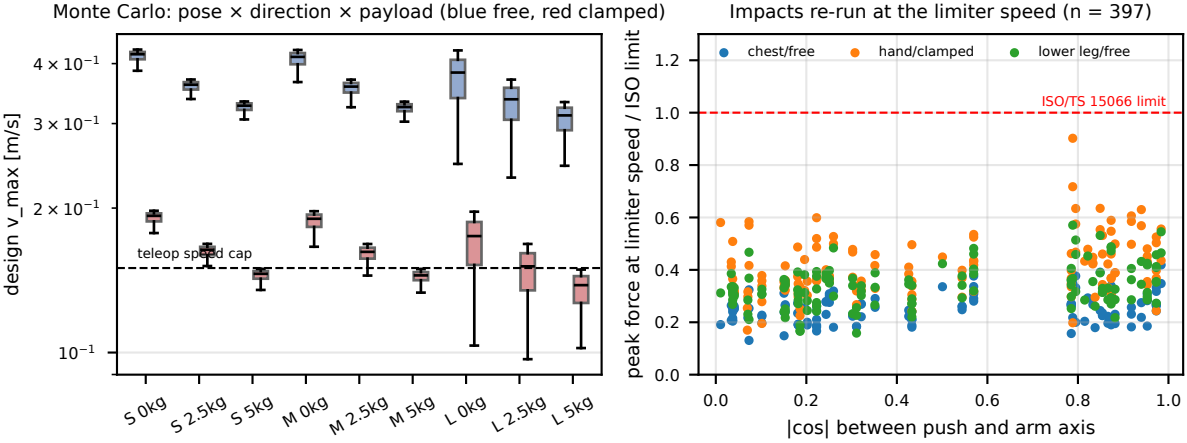
Monte Carlo: pose × direction × payload (blue free, red clamped)
design v_max [m/s]
4 × 10⁻¹
3 × 10⁻¹
2 × 10⁻¹
10⁻¹
teleop speed cap
S 0kg
S 2.5kg
S 5kg
M 0kg
M 2.5kg
M 5kg
L 0kg
L 2.5kg
L 5kg
Impacts re-run at the limiter speed (n = 397)
chest/free
hand/clamped
lower leg/free
ISO/TS 15066 limit
peak force at limiter speed / ISO limit
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
|cos| between push and arm axis